Standard operating procedures for pre-registered experiments

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Tags: [G] adopted from Lin et al. (2016), [A] adapted from it, [N] new here.

1. STATUS AND SCOPE

1.1 Purpose. This document sets default practices for analytic decisions that a pre-analysis plan (PAP) leaves open. It follows the model of Lin et al. (2016); Lin and Green (2016) give the rationale. [G]

1.2 Precedence. The PAP overrides this document. [G] This document overrides any decision taken after data collection began. [N]

1.3 Binding force. A version binds a study only if that study's PAP cites that version by number and address and the registration holds a copy of it. A later version never applies retroactively. Using a later rule is a deviation under section 8. [N]

1.4 Scope. Randomized field, audit, and survey experiments, including multi-arm designs and megastudies. Observational studies and conjoint experiments are not covered in this version. [N]

1.5 Adoption. Adoption is decided per project with all co-authors, before registration. [N]

1.6 Gaps. Where this document is silent, Lin et al. (2016, version 1.05) applies, including its rules on block randomization, treatment/placebo designs, and duplicate records. Its block randomization rules cover two arms only, so the PAP of a block-randomized multi-arm design states its own. Its section on voter turnout and the voter turnout defaults elsewhere in it do not apply. [A]

2. BEFORE REGISTRATION

2.1 The PAP states, for every test: the outcome, how that outcome is measured in each arm, the reference arm, the primary specification, and the multiple-testing family. [N]

2.2 Randomization code is checked by at least two team members and made public, with the seed recorded. [A]

2.3 A dated deviation log opens on the day of registration (section 8). [N]

3. DATA PREPARATION

3.1 All steps from raw data to analysis files run in code, and the code is made public. [G]

3.2 Before dissemination or submission the team documents that treatment occurred, that outcome data were gathered, that at least two team members were involved in all raw data collection, that the code correctly implements the intended logic and reproduces the analysis data from the raw data, and that the distributions and missingness of all analysis variables and their precursors were checked for plausibility and for consistency across processing stages, normally with cross-tabulations as well as univariate summaries. Any unresolved issue is disclosed. [G]

3.3 If an outcome is established by a different procedure in some arms (for example, a manual check for the control group), the Methods section describes that procedure, the team names who carried it out, and the record is kept. A missing value does not count as a zero until this is done. [N]

3.4 Non-answer codes ("don't know", "prefer not to say") are recoded to missing before any count. The value tally of a variable is inspected before filtering on it. [N]

3.5 Dates and durations in the text are read from timestamps in the data. They are never computed from a schedule index such as "day 77". [N]

4. ESTIMATION

4.1 For random assignment of individuals, with M the size of the smallest arm and N the sample size: if M is at least 20, regress the outcome on treatment, mean-centered covariates, and their interactions (Lin, 2013). If M is below 20 and N is at least 20, drop the interactions. If N is below 20, use a difference in means or a difference in differences. [G]

4.2 In a multi-arm design, each arm has its own indicator and interactions, and M refers to the smallest arm. [N]

4.3 Covariates are measured before randomization, unless there is a convincing argument that a variable measures pre-randomization conditions and treatment did not affect its measurement, or that treatment could not have affected it. The model holds at most M/20 covariates with interactions and at most N/20 without. If the PAP names none and at most 10 baseline covariates exist within these limits, not counting transformations and interactions between covariates, all enter. [G]

4.4 Missing covariates: at most 10% missing, recode to the overall mean, never an arm-specific mean. Above 10%, add a missingness dummy and recode to a constant, reverting to the mean if the dummies breach the limit in 4.3. [G]

4.5 Unadjusted estimates are always reported next to the adjusted ones, and the text states which estimator was pre-specified as primary. [G]

4.6 Least squares is the default for binary and count outcomes. Supplementary estimates from nonlinear models, using marginal effects, are supplied if referees ask for them, with logit preferred to probit. [G]

4.7 A regression that drops a collinear term stops with an error. Before analyzing actual outcomes, the team should attempt the permutation test on mock outcome data and the actual covariates, and drops covariates until it runs without errors. [A] Any term removed is named in the report. [N]

5. INFERENCE

5.1 Standard errors are HC2 without clustering and Bell and McCaffrey's bias-reduced estimator with clustering, with Bell-McCaffrey degrees of freedom also with clustering, not the degrees of freedom of Imbens and Kolesár (2016). [G] The analysis script names the package and version. [N]

5.2 Standard errors are clustered if and only if a regression holds several observations for some randomization units, and then at the level of the randomization unit, not higher. [G]

5.3 Tests are two-tailed unless the PAP specifies otherwise. [G]

5.4 p-values come from a Studentized permutation test: the t-statistic is compared with its distribution under random reassignments that follow the actual randomization scheme, with 10,000 randomizations and the random number seed 1234567. Two-sided values double the smaller one-sided value, capped at 1. Confidence intervals follow rule 5.1. [G]

5.5 For a contrast between two arms of a multi-arm design, only subjects in those two arms are reassigned. The number of randomizations rises where needed so that the smallest attainable p-

value lies below .05 divided by the family size. [N]

6. MULTIPLE TESTING

Lin et al. (2016, version 1.05) has no rule on this, so the whole section is new. [N]

6.1 The PAP defines the families. If it does not, rules 6.2 to 6.4 apply and the report discloses the gap.

6.2 A family holds registered tests only: one family per registered outcome within a research question. The family of the registered primary outcome is confirmatory. Both specifications enter if the PAP registers both.

6.3 These never enter a family: unregistered outcomes, exploratory contrasts, intercepts, covariate coefficients, and contrasts against a reference arm chosen after results were seen.

6.4 The default procedure is Benjamini and Hochberg (1995), which requires no estimate of the share of true nulls. If the PAP names Storey and Tibshirani (2003), that procedure is primary, the estimate of the share of true nulls is reported, Benjamini-Hochberg values appear alongside, and the share is fixed at 1 in families below 20 tests.

6.5 Every test is reported with its estimate, unadjusted 95% interval, unadjusted p-value, adjusted value, and family size.

6.6 False discovery rate procedures define no simultaneous intervals, so plotted intervals are unadjusted 95% intervals by default. A familywise-adjusted interval names its procedure and the number of estimates. The script confirms the label by comparing (estimate minus lower bound) / SE with the expected critical value, because software can substitute a procedure without an error.

6.7 If an arm is selected as best on its estimates, the report states that this estimate is subject to a winner's curse and gives a hybrid estimate and interval (Andrews et al., 2024) or labels the conventional one descriptive. A large sample does not exempt it. Shrinkage can reduce the curse but does not correct it in general, so it does not replace this step. A contrast the PAP names and reports whatever its result is not selected in this sense; one reported or headlined because of its result is. A difference between the winner and the runner-up stays descriptive.

6.8 A sensitivity check may pool families, and is labeled as such.

7. OUTCOMES, EXCLUSIONS, AND EVENTS DURING THE STUDY

7.1 As a rule, outcomes are not coded in ways that exclude subjects from the estimation of treatment effects, and subjects with an outcome of zero stay in. Exceptions are exclusions that do not threaten symmetry between arms, and subgroups identified after assignment whose outcomes were determined before treatment began, if treatment assignment could not plausibly have affected outcomes before the actual treatment began and comparability checks find no problem (Lin et al., 2016, version 1.05, p. 22). [G]

7.2 Outcomes defined only for those who respond (tone, length, quality of a reply in an audit study) are not analyzed conditional on response, because response is post-treatment (Coppock, 2019; Montgomery et al., 2018). The effect on response is reported first. The default for quality is the redefined outcome, equal to the measure if the subject responded and a value fixed in the PAP otherwise, labeled as a composite of response and quality and never as an effect on quality. Trimming bounds among always-responders (Lee, 2009) follow, with the case for monotonicity stated. An always-responder subgroup is a sensitivity analysis only. [N]

7.3 Attrition (missing outcomes): the team checks symmetric implementation, compares attrition rates across arms, and compares attrition patterns across arms by treatment-covariate interactions, with p below .05 in either comparison as evidence of asymmetry. After a red flag from any of these checks and without a PAP rule: second-round sampling with extreme value bounds if resources and assumptions allow. Otherwise a disinterested jury decides whether monotonicity is plausible: if so, trimming bounds (Lee, 2009); if not, extreme value bounds. The PAP analysis is always reported as well. [G]

7.4 Except as the PAP specifies or in a supplemental robustness check, outliers are not removed or edited merely for being extreme. They may trigger data integrity checks if the process and any resulting edits are results-blind and symmetric across arms. [G]

7.5 With noncompliance, the report includes a test of the hypothesis that the average intent-to-treat effect is zero. [G]

7.6 Covariate balance is tested jointly with a permutation test. A p-value of .01 or lower prompts a review of assignment and data handling. If the review finds no error, the imbalance is reported, with estimates with and without adjustment. [G]

7.7 Survey experiments: subjects who discern the purpose are not excluded. An analysis excluding respondents who fail screener questions is reported if treatment cannot plausibly affect attention and the comparability checks of Lin et al. (2016) pass, with full-sample results alongside. [G]

7.8 A decision needing judgment after data collection began (a new covariate source, a monotonicity assumption) goes to colleagues outside the team, who see no estimates and no information on the direction of any effect or imbalance. [A]

7.9 A canceled, stopped, or failed study receives a public summary of design, implementation, any results, and apparent causes. [G]

8. DEVIATIONS AND REPORTING

8.1 Each article references its PAP and, where the PAP adopts this document, the version adopted. Analyses that deviate from the PAP are marked as not pre-specified. Omitted pre-specified analyses are described and made available in a document the article references. [A]

8.2 Each deviation, and each use of a default of this document where the PAP is silent, receives a log entry. The entry records: the date; whether outcome data were available or seen; the PAP text and code line at issue; what was done; the reason; the flexibility the change left and its effect on the severity of the test and, where relevant, on validity; and the result under the registered analysis, or why that analysis cannot be run or no longer makes sense. The reason names one or more categories in Lakens (2024): unforeseen event, mistake in the preregistration or study, missing information, violated untested assumption, or falsified auxiliary hypothesis. Any other reason is described. [N]

8.3 The word "pre-registered" is used only for what the PAP states. A rule chosen after a result was known is stated as such, next to the alternative it replaced. Where the PAP was silent, the report states the other plausible analyses and whether they support the same claim. [N]

8.4 The report lists which rules of this document were invoked. [N]

9. CODE AND CLAIM CHECKS

9.1 Every number in a manuscript traces to a script output. Countable claims are asserted in code, so that a change in the data fails the script. [N]

9.2 Scripts assert the expected size of each family (6.2) and of each estimation sample. Any resampling is checked against the estimation sample first, since fitted values and residuals exclude dropped rows. [N]

9.3 A second team member re-derives the headline estimates from the analysis data. [N]

9.4 Generative AI tools may write code and text. A named team member is responsible for each number. Agreement between two reviewers or agents who saw the same files does not count as an independent check. [N]

9.5 A claim that data, code, or documentation does not exist states where the team stores such material and what was searched, and is confirmed with the responsible team member before it enters a report. [N]

10. VERSIONING

10.1 Versions 0.x are drafts and cannot be cited in a PAP. Version 1.0 is the first citable version. Each version is published on the author's homepage at its own address, and its file is never replaced. [N]

10.2 Each release is frozen, and changes go into a new version with an entry in the change log. [N]

Change log

- 1.0 (2026-11-04): first citable version.

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